

Context Free Grammars.

$$G = (N, \Sigma, P, S).$$

N - finite set (non-terminal symbols).

Σ - finite set (terminal symbols)

Assumption : $N \cap \Sigma = \emptyset$.

P - finite subset of $N \times (N \cup \Sigma)^*$ [productions]

Language generated by G .

$$L(G) = \{x \in \Sigma^* \mid S \xrightarrow[G]{*} x\}$$

$B \subseteq \Sigma^*$ is a context free language (CFL) if

$B = L(G)$ for some CFG G .

Balanced Parenthesis.

$G = (N, \Sigma, P, S)$ where

$$\Sigma = \{[,]\} \quad N = \{S\}$$

$$P: S \rightarrow [S] \mid SS \mid \epsilon$$

Claim. $L(G) = \{x \in \{[,]\}^* \mid x \text{ is balanced}\}$.

$L(x) = \# [(x)$: Number of $[$ in string x .

$R(x) = \#] (x)$: Number of $]$ in string x .

$x \in \Sigma^*$ is balanced iff

1. $L(x) = R(x)$

2. For all prefix y of x ; $L(y) \geq R(y)$

$$G: S \rightarrow [S] \mid SS \mid \epsilon.$$

x is balanced iff
 (1) $L(x) = R(x)$
 (2) $L(y) \geq R(y) \neq \text{prefix}_y \text{ of } x.$

Theorem. $L(G) = \{x \in \{[,.]\}^* \mid x \text{ satisfies (1) and (2)}\}$

Claim. if $S \xrightarrow{*}_G x$ then x satisfies (1) and (2)

Proof.

Induction on the length of the derivation

For any $\alpha \in (N \cup \Sigma)^*$, if $S \xrightarrow{*}_G \alpha$ then α satisfies (1.) and (2.)

Base case: if $S \xrightarrow{0}_G \alpha$ then $\alpha = S$

$$G: S \rightarrow [S] \mid SS \mid \epsilon. \quad \left| \begin{array}{l} x \text{ is balanced iff} \\ (1) L(x) = R(x) \\ (2) L(y) \geq R(y) \text{ } \forall \text{ prefix } y \text{ of } x. \end{array} \right.$$

Theorem. $L(G) = \{x \in \{[,]\}^* \mid x \text{ satisfies (1) and (2)}\}$

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Induction on the length of the derivation

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Base case: if $S \xrightarrow{0}_G \alpha$ then $\alpha = S$

Induction step. Suppose $S \xrightarrow{n+1}_G \alpha \mid S \xrightarrow{n}_G \beta \xrightarrow{1}_G \alpha$

By induction hypothesis, β satisfies (1) and (2).

$S \rightarrow \epsilon, S \rightarrow SS \mid \exists \beta_1, \beta_2 \in (N \cup \Sigma)^* \text{ s.t.}$

$\beta = \beta_1 S \beta_2 \quad \alpha = \begin{cases} \beta_1 \beta_2 & \text{if } S \rightarrow \epsilon \\ \beta_1 SS \beta_2 & \text{if } S \rightarrow SS \end{cases}$

if $S \rightarrow [S]$ $\exists \beta_1, \beta_2 \in (N \cup \Sigma)^*$, $\beta = \beta_1 S \beta_2$

$$\alpha = \beta_1 [S] \beta_2.$$

By IH, β satisfies (1) & (2)

$$L(\alpha) = L(\beta) + 1 = R(\beta) + 1 = R(\alpha)$$

Thus α satisfies (1.)

$$S \xrightarrow[n]{G} \overbrace{\beta_1 S \beta_2}^{\beta} \xrightarrow[G]{1} \beta_1 [S] \beta_2 = \alpha$$

To show: α satisfies 2.

Let γ be an arbitrary prefix of α .

To show: $L(\gamma) \geq R(\gamma)$

— γ is a prefix of $\beta_1 \Rightarrow \gamma$ is a prefix of β

— γ is a prefix of $\beta_1 [S]$ but not of β_1

$$L(\gamma) = L(\beta_1) + 1 \geq R(\beta_1) + 1 \quad [\text{IH, Since } \beta_1 \text{ is a prefix of } \beta]$$

$$\downarrow$$

$$> R(\beta_1) = R(\gamma)$$

— $\gamma = \beta_1 [S] \delta$; where δ is a prefix of β_2 .

$$S \xrightarrow{G} \overbrace{\beta_1 S \beta_2}^{\beta} \xrightarrow{G} \beta_1 [S] \beta_2 = \alpha$$

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$$\begin{aligned} L(\gamma) = L(\beta_1) + 1 &\geq R(\beta_1) + 1 \quad [\text{IH, Since } \beta_1 \text{ is a prefix of } \beta] \\ &\downarrow \\ &> R(\beta_1) = R(\gamma) \end{aligned}$$

— $\gamma = \beta_1 [S] \delta$; where δ is a prefix of β_2 .

$$L(\gamma) = L(\beta_1 S \delta) + 1$$

$$\geq R(\beta_1 S \delta) + 1 \quad [\text{Induction hypothesis}]$$

$$= R(\gamma).$$

$$S \xrightarrow[G]{n} \overbrace{\beta_1 S \beta_2}^{\beta} \xrightarrow[G]{1} \beta_1 [S] \beta_2 = \alpha$$

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— $\gamma = \beta_1 [S] S$; where S is a prefix of β_2 .

$$L(\gamma) = L(\beta_1 S S) + 1$$

$$\geq R(\beta_1 S S) + 1 \quad [\text{Induction hypothesis}]$$

$$= R(\gamma).$$

Thus, in all the cases, we have shown that

$$L(\gamma) \geq R(\gamma).$$

Therefore, α satisfies (2.)

claim. if x satisfies (1) and (2) then $S \xrightarrow{*}_G x$

Proof. Induction on $|x|$.

Base Case. $|x|=0$ - Trivial since $S \rightarrow \epsilon \leadsto P$

Induction step. if $|x|>0$ - Consider 2 cases.

(a) $\exists$ a proper prefix y of x satisfying (1) ~~and~~ (2).
 $(0 < |y| < |x|)$.

(b) no such prefix exists.

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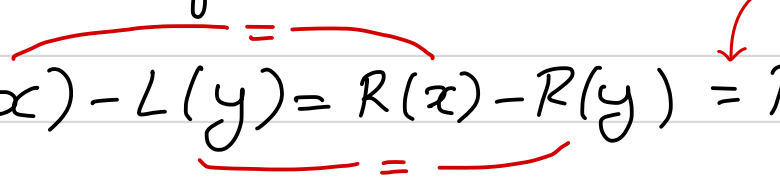
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Case (a). $x = yz$ for some z , $0 < |z| < |x|$.

Claim. z satisfies (1) and (2)

$$L(z) = L(x) - L(y) = R(x) - R(y) = R(z)$$



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$$L(z) = L(x) - L(y) = R(x) - R(y) = R(z)$$

By Definition (arrow pointing to $R(y)$)

For any prefix w of z

$$L(w) = L(yw) - L(y)$$

$$\geq R(yw) - R(y) \quad \left(\begin{array}{l} \text{Since } yw \text{ is a prefix of } x \\ \text{and } L(y) = R(y) \end{array} \right)$$
$$= R(w). \quad (\text{By definition})$$

By induction hypothesis $S \xrightarrow{*}_G y$, $S \xrightarrow{*}_G z$

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 $\rightarrow (0 < |y| < |x|)$.

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By induction hypothesis $S \xrightarrow{*}_G y$, $S \xrightarrow{*}_G z$

$$S \xrightarrow{1}_G SS \xrightarrow{*}_G yS \xrightarrow{*}_G yz = x.$$

Case b. $x = [z]$ for some z .

Claim: z satisfies (1) and (2).

$$L(z) = L(x) - 1 = R(x) - 1 = R(z).$$

$\forall$ non-null prefixes u of z ,

$$L(u) - R(u) = L([u]) - 1 - R([u]) \geq 0$$

$$\text{Since } L([u]) - R([u]) \geq 1.$$

By induction hypothesis $S \xrightarrow[G]{*} z$

Case b. $x = [z]$ for some z .

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$\forall$ non-null prefixes u of z ,

$$L(u) - R(u) = L([u]) - 1 - R([u]) \geq 0$$

$$\text{Since } L([u]) - R([u]) \geq 1.$$

By induction hypothesis $S \xrightarrow[G]{*} z$

$$S \xrightarrow[G]{1} [S] \xrightarrow[G]{*} [z] = x.$$

$$G = (N, \Sigma, P, S)$$

$$S \rightarrow SS \mid 01 \mid 11 \in.$$

A "long" derivation of 01 in G :

$$S \xrightarrow[G]{1} SS \xrightarrow[G]{1} SSS \xrightarrow[G]{1} SSSS \xrightarrow[G]{1} SSSS \xrightarrow[G]{1} SSS \xrightarrow[G]{1} SS \rightarrow OS \rightarrow 01$$

Chomsky Normal Form. (CNF)

$G = (N, \Sigma, P, S)$ is in CNF if all productions

$$A \rightarrow BC \quad A \rightarrow a \quad A, B, C \in N, a \in \Sigma.$$